

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

7 Logic and Proof (mj201)

Decision
procedures

- (a) Exhibit a model for the following set of formulas, or prove that none exists. Briefly explain your work in each step.

$$P \quad P \rightarrow (R \rightarrow Q) \quad P \vee \neg Q \vee \neg P \quad Q \rightarrow S \wedge \neg T \quad S \rightarrow Q \vee T$$

[6 marks]

Answer: There are a number of ways to answer this, but DPLL is clearly invited to be used, and other ad hoc methods will not get full marks.

The first step is to convert the formulas into clauses. In this case, it is all but trivial.

$$\{P\} \quad \{\neg P, \neg R, Q\} \quad \{P, \neg Q, \neg P\} \quad \{\neg Q, S\} \quad \{\neg Q, \neg T\} \quad \{\neg S, Q, T\}$$

Tautological elimination for P :

$$\{P\} \quad \{\neg P, \neg R, Q\} \quad \{\neg Q, S\} \quad \{\neg Q, \neg T\} \quad \{\neg S, Q, T\}$$

Unit P , so P is true:

$$\{\neg R, Q\} \quad \{\neg Q, S\} \quad \{\neg Q, \neg T\} \quad \{\neg S, Q, T\}$$

Pure $\neg R$, so R is false:

$$\{\neg Q, S\} \quad \{\neg Q, \neg T\} \quad \{\neg S, Q, T\}$$

There are no unit clauses and no pure literals, so a case split is forced. For example, we can try if Q is true (unit):

$$\{S\} \quad \{\neg T\}$$

Now unit S (also pure) so S is true:

$$\{\neg T\}$$

Now unit $\neg T$ (also pure) so T is false:

we have all clauses deleted.

Case split for Q is false is not needed.

The original set of clause is satisfiable as no clauses are left (all deleted). No need to do the second branch of the case split. Other runs of the methods are possible. The model is: $P = \text{true}$, $Q = \text{true}$, $R = \text{false}$, $S = \text{true}$, $T = \text{false}$.

Modal logic

- (b) For each of the following sets of formulas, either exhibit an interpretation in S4 modal logic that satisfies them simultaneously at a particular world, w , or show through a formal proof that they cannot be satisfied.

(i) $\Diamond \Box P, \quad Q, \quad \Box \Diamond \Box \neg Q, \quad \Box(P \rightarrow \Diamond R \wedge \Diamond \neg R), \quad \Box(\Box \neg Q \vee \neg \Diamond P)$

[8 marks]

Answer: Note that the formulas $\Diamond \Box P$, Q , and $\Box(\Box \neg Q \vee \neg \Diamond P)$ cannot be satisfied simultaneously for a world w . Below is a sequent calculus proof that they are contradictory.

$$\begin{array}{c}
 \frac{\overline{\Diamond\Box P, Q \Rightarrow Q} \text{ (basic)}}{\overline{\Diamond\Box P, Q, \neg Q \Rightarrow} \text{ (}\neg l\text{)}} \quad \frac{\overline{P \Rightarrow P} \text{ (basic)}}{\overline{\Box P \Rightarrow P} \text{ (}\Box l\text{)}} \quad \frac{\overline{\Box P \Rightarrow \Diamond P} \text{ (}\Diamond r\text{)}}{\overline{\Diamond\Box P, Q \Rightarrow \Diamond P} \text{ (}\Diamond l\text{)}} \\
 \frac{\overline{\Diamond\Box P, Q, \Box\neg Q \Rightarrow} \text{ (}\Box l\text{)}}{\overline{\Diamond\Box P, Q, \Box\neg Q \vee \neg\Diamond P \Rightarrow} \text{ (}\vee l\text{)}} \quad \frac{\overline{\Diamond\Box P, Q, \neg\Diamond P \Rightarrow} \text{ (}\neg l\text{)}}{\overline{\Diamond\Box P, Q, \Box(\Box\neg Q \vee \neg\Diamond P) \Rightarrow} \text{ (}\Box l\text{)}}
 \end{array}$$

(ii) $\Box(P \vee Q), \Diamond\neg P, \neg\Diamond Q$

[6 marks]

Answer: The set of formulas cannot be satisfied. We show this by proving that $\Box(P \vee Q), \Diamond\neg P, \neg\Diamond Q$ entails a contradiction

$$\begin{array}{c}
 \frac{\overline{Q \Rightarrow P, Q} \text{ (basic)}}{\overline{P \vee Q \Rightarrow P, Q} \text{ (}\vee l\text{)}} \quad \frac{\overline{P \Rightarrow P, Q} \text{ (basic)}}{\overline{\Box(P \vee Q) \Rightarrow P, Q} \text{ (}\Box l\text{)}} \\
 \frac{\overline{\Box(P \vee Q) \Rightarrow P, Q} \text{ (}\Box l\text{)}}{\overline{\Box(P \vee Q) \Rightarrow P, \Diamond Q} \text{ (}\Diamond r\text{)}} \\
 \frac{\overline{\Box(P \vee Q), \neg P \Rightarrow, \Diamond Q} \text{ (}\neg l\text{)}}{\overline{\Box(P \vee Q), \Diamond\neg P \Rightarrow \Diamond Q} \text{ (}\Diamond l\text{)}} \\
 \frac{\overline{\Box(P \vee Q), \Diamond\neg P, \neg\Diamond Q \Rightarrow} \text{ (}\neg l\text{)}}{}
 \end{array}$$
